of living things and the different types of reproduction have also been discussed. Lower plants, fungi and animals show different methods of asexual reproduction. Some of the methods include, formation of Conidia (*Aspergillus* and *Penicillium*); Budding (*Yeast* and *Hydra*); Fragmentation (*Spirogyra*); production of Gemma (*Marchantia*); Regeneration (*Planaria*) and Binary fission (Bacteria) (Refer chapter 1 of Unit I of class XI). The individuals formed by this method is morphologically and genetically identical and are called **clones**. Higher plants also reproduce asexually by different methods which are given below:

1.2 Vegetative reproduction

1.2.1 Natural methods

Natural vegetative reproduction is a form of asexual reproduction in which a bud grows and develops into a new plant. The buds may be formed in organs such as root, stem and leaf. At some stage, the new plant gets detached from the parent plant and starts to develop into a new plant. Some of the organs involved in the vegetative reproduction also serve as the organs of storage and perennation. The unit of reproductive structure used in propagation is called **reproductive propagules** or **diaspores.** Some of the organs that help in vegetative reproduction are given in Figure 1.1.

A. Vegetative reproduction in root

The roots of some plants develop vegetative or adventitious buds on them. Example *Murraya, Dalbergia* and *Millingtonia*. Some tuberous adventitious roots apart from developing buds also store food. Example *Ipomoea batatus* and *Dahlia*. Roots possessing buds become detached from the parent plant and grow into independent plant under suitable condition.

Activity

Visit to a vegetable market and classify the vegetables into root, stem or leaf based on their utility and identify how many of them can be propagated through asexual methods.



Scourge of water bodies / Water hyacinth (*Eichhornia crassipes)* is an invasive weed on water bodies like ponds, lakes and reservoirs. It is popularly called "Terror of Bengal". It spreads rapidly through offsets all over the water body and depletes the dissolved oxygen and causes death of other aquatic organisms.



B. Vegetative reproduction in stem

From the Unit 3 of class XI (Vegetative morphology) you are familiar with the structure of various underground stem and sub aerial stem modifications. These include rhizome (*Musa paradisiaca*, *Zingiber officinale* and *Curcuma longa*); corm (*Amorphophallus* and *Colocasia*); tuber (*Solanum tuberosum*); bulb(*Allium cepa* and *Lilium*) runner (*Centella asiatica*); stolon (*Mentha,* and *Fragaria*); offset (*Pistia,* and *Eichhornia*); sucker (*Chrysanthemum*) and bulbils (*Dioscorea* and *Agave*). The axillary buds from the nodes of rhizome and eyes of tuber give rise to new plants.

C. Vegetative reproduction in leaf

In some plants adventitious buds are developed on their leaves. When they are detached from the parent plant they grow into new individual plants. Examples: *Bryophyllum, Scilla,* and *Begonia*. In *Bryophyllum*, the leaf is succulent and notched on its margin. Adventious buds develop at these notches and are called **epiphyllous buds**. They develop into new plants forming a root system and become independent plants when the leaf gets decayed. *Scilla* is a bulbous plant and grows in sandy soils. The foliage leaves are long and narrow and epiphyllous buds develop at their tips. These buds develop into new plants when they touch the soil.

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